Diploma Programme
Programme du diplôme
Programa del Diploma

Chemistry
Higher level
Paper 2

Tuesday 12 November 2024 (afternoon)

2 hour 15 minutes

Candidate session number

Instructions to candidates

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **chemistry data booklet** is required for this paper.
- The maximum mark for this examination paper is **[95 marks]**.

Answer **all** questions. Answers must be written within the answer boxes provided.

1. A student determined the percentage of the active ingredient iron(III) hydroxide, $Fe(OH)_3$, in a 1.50 g antacid preparation.

The preparation was dissolved in 40.00 cm^3 of 0.150 mol dm^{-3} hydrochloric acid, which was in excess.

(a) Calculate the amount, in mol, of HCl. [1]

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(b) Formulate the equation for the reaction of HCl with $Fe(OH)_3$. [1]

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(c) The excess hydrochloric acid required 18.40 cm^3 of 0.1205 mol dm^{-3} NaOH for neutralization. Calculate the amount of excess acid present. [1]

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(d) Calculate the amount of HCl that reacted with $Fe(OH)_3$. [1]

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Turn over

(Question 1 continued)

(e) Determine the mass of $Fe(OH)_3$ in the antacid preparation. [2]

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(f) Calculate the percentage by mass of iron(III) hydroxide in the 1.50 g antacid preparation to three significant figures. [1]

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(g) Outline why repeating quantitative measurements is important. [1]

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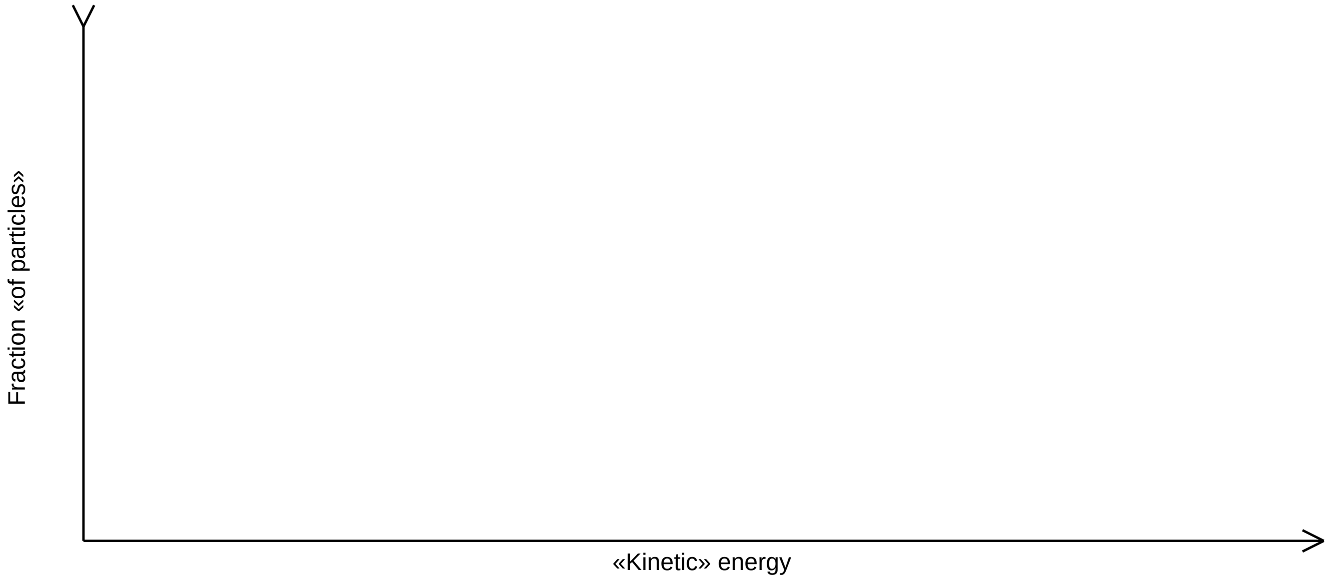
2. Data analysis is a fundamental tool in the study of chemical kinetics.

(a) Sketch a Maxwell−Boltzmann distribution curve for a gas-phase reaction showing the activation energies with and without a homogeneous catalyst. [3]



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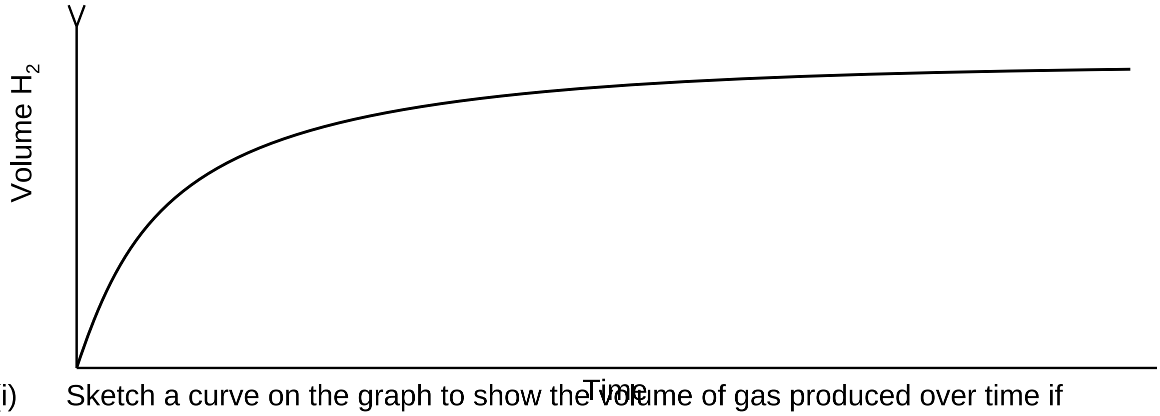
(Question 2 continued)

(b) Excess nitric acid is added to lumps of zinc oxide. The graph shows the volume of hydrogen gas, H_2, produced over time.



(i) Sketch a curve on the graph to show the volume of gas produced over time if the same mass of powdered zinc oxide is used instead of lumps. All other conditions remain constant. [1]

(ii) State and explain the effect on the rate of reaction if propanoic acid of the same concentration is used in place of nitric acid. [2]

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(c) Outline why pH is more widely used than $[H^+]$ for measuring relative acidity. [1]

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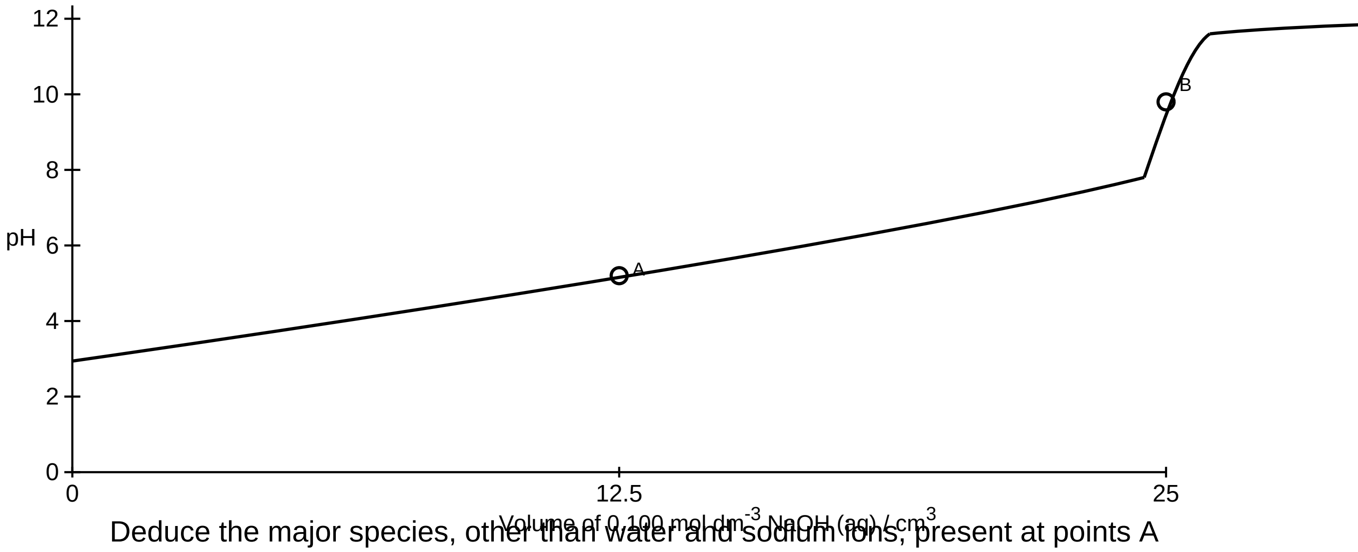
(Question 2 continued)

(d)

(i) The graph represents the titration of 25.00 cm^3 of 0.100 mol dm^{-3} aqueous propanoic acid with 0.100 mol dm^{-3} aqueous sodium hydroxide.



Deduce the major species, other than water and sodium ions, present at points A and B during the titration. [2]

A:

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B:

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(This question continues on the following page)

(Question 2 continued)

(ii) Calculate the pH of 0.100 mol dm^{-3} aqueous propanoic acid.
$K_a = 1.34 \times 10^{-5}$ [2]

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(iii) Outline, using an equation, why sodium propanoate is basic. [1]

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(iv) Predict whether the pH of an aqueous solution of methylammonium chloride will be greater than, equal to or less than 7 at 298 K. [1]

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(e)

(i) Formulate the equation for the reaction of sulfur trioxide, SO_3, with water to form an acid. [1]

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(ii) Formulate the equation for the reaction of the acid formed in (e)(i) with magnesium carbonate. [1]

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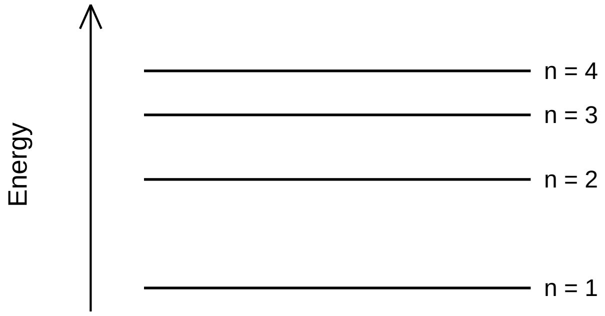
Turn over

3. The emission spectrum of hydrogen provides evidence for the quantised nature of energy.

(a)

(i) Draw the first four energy levels of a hydrogen atom on the axis, labelling $n = 1, 2, 3$ and 4. [1]



(ii) Draw the lines, on your diagram, that represent the electron transitions to $n = 2$ in the emission spectrum. [1]

(iii) Hydrogen spectral data give the frequency of 6.17×10^{14} s^{-1} for the H_α line in the Balmer series.
Calculate the energy, in J, for a single photon of this radiation using section 1 of the data booklet. [1]

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(iv) Calculate the wavelength, in m, of the H_α line using section 1 of the data booklet. [1]

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(This question continues on the following page)

(Question 3 continued)

(b) Elements show trends in their properties across the periodic table.

(i) Outline why atomic radius decreases across period 2, lithium to fluorine. [1]

(ii) Outline why the ionic radius of Na^+ is smaller than that of F^-. [2]

(c)

(i) Iron is widely used as a structural material.
Draw arrows in the boxes to represent the electronic configuration of iron in the 4s and 3d orbitals. [1]

4s 3d

(ii) Iron can be electroplated onto steel. In the electrolytic cell, an iron anode (positive electrode) is used with iron(II) sulfate solution and a steel cathode (negative electrode).
Formulate the half-equation at each electrode. [2]

Anode (positive electrode):

Cathode (negative electrode):

(This question continues on the following page)

(Question 3 continued)

(iii) Outline where and in which direction the electrons flow during electroplating. [1]

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(iv) Deduce any change in the concentration of the electrolyte during electroplating. [1]

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(v) Deduce the gas formed at the anode (positive electrode) when platinum is used in place of iron. [1]

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(d) Explain why transition metals exhibit variable oxidation states in contrast to alkali metals. [2]

Transition metals:

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. . . .

Alkali metals:

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. . . .

Turn over

4.

(a) In acidic solution, permanganate ions, MnO_4^- (aq), oxidize iron(II) ions, Fe^{2+} (aq).

$MnO_4^-\ (aq) + 8H^+\ (aq) + 5e^- \rightleftharpoons Mn^{2+}\ (aq) + 4H_2O\ (l)$
$Fe^{2+}\ (aq) \rightleftharpoons Fe^{3+}\ (aq) + e^-$

Formulate the equation for the redox reaction. [1]

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(b) The change in the Gibbs free energy for the reaction under standard conditions, ΔG^{o}, is −419 kJ at 298 K.
Determine the value of E^{o}, in V, for the reaction using sections 1 and 2 of the data booklet. [2]

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(c) Calculate the standard electrode potential, in V, for the MnO_4^-/Mn^{2+} reduction half-equation using section 24 of the data booklet. [1]

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Turn over

5. Enthalpy changes depend on the number and type of bonds broken and formed.

(a) Ammonia is manufactured industrially by the Haber process.

$N_2\ (g) + 3H_2\ (g) \longrightarrow 2NH_3\ (g)$

Determine the enthalpy change, ΔH, for the reaction, in kJ, using section 11 of the data booklet.

Bond enthalpy for N≡N: 945 kJ mol^{-1} [3]

(b) The table lists the standard enthalpies of formation, ΔH^o_f, for some species.

	N_2 (g)	**H_2 (g)**	**NH_3 (g)**
ΔH^o_f / kJ mol^{-1}	—	—	−46.0

(i) Outline why no value is listed for N_2 (g) and H_2 (g). [1]

(ii) Determine the value of ΔH^o, in kJ, for the reaction using the values in the table. [1]

(This question continues on the following page)

(Question 5 continued)

(iii) Outline why the value of enthalpy of reaction calculated from bond enthalpies may differ from the value using standard enthalpies of formation. [1]

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(c) The table lists standard entropy, S^{o}, values.

	N_2 (g)	H_2 (g)	NH_3 (g)
S^{o} / J K^{-1} mol^{-1}	+192	+131	+193

Calculate the standard entropy change for the reaction, ΔS^{o}, in J K^{-1}.

$N_2 (g) + 3H_2 (g) \longrightarrow 2NH_3 (g)$ [1]

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(d) Calculate the standard Gibbs free energy change, ΔG^{o}, in kJ, for the reaction at 298 K using your answer to (b)(ii). [1]

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(e) Determine the temperature, in K, above which the reaction becomes non-spontaneous. [1]

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Turn over

6. A mixture of 2.00 mol N_2 (g), 3.00 mol H_2 (g) and 1.00 mol NH_3 (g) is placed in a 2.00 dm^3 container and allowed to reach equilibrium.

$N_2 (g) + 3H_2 (g) \rightleftharpoons 2NH_3 (g)$

(a) Distinguish between the terms reaction quotient, Q_c, and equilibrium constant, K_c. [1]

(b) The equilibrium constant, K_c, is 6.02×10^{-2} at temperature T.
Deduce, showing your work, the direction of the initial reaction. [2]

(c)

(i) Dinitrogen tetroxide is in equilibrium with nitrogen dioxide.

$N_2O_4 (g) \rightleftharpoons 2NO_2 (g) \quad \Delta H^o > 0$

Deduce, giving a reason, the effect of increasing the temperature on the concentration of N_2O_4. [1]

(This question continues on the following page)

(Question 6 continued)

(ii) A two-step mechanism is proposed for the formation of NO_2 (g) from NO (g) that involves an endothermic equilibrium process.

First step: $2NO\ (g) \rightleftharpoons N_2O_2\ (g)$ fast

Second step: $N_2O_2\ (g) + O_2\ (g) \longrightarrow 2NO_2\ (g)$ slow

Deduce the rate expression for the mechanism. [2]

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(d) The rate constant for a reaction triples when the temperature is increased from 20.0 °C to 30 °C.
Calculate the activation energy, E_a, in kJ mol^{-1} for the reaction using sections 1 and 2 of the data booklet. [2]

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Turn over

7. Some physical properties of molecular substances result from different types of intermolecular forces.

(a)

(i) Explain why the hydrides of group 17 elements (HF, HCl, HBr and HI) are polar molecules. [2]

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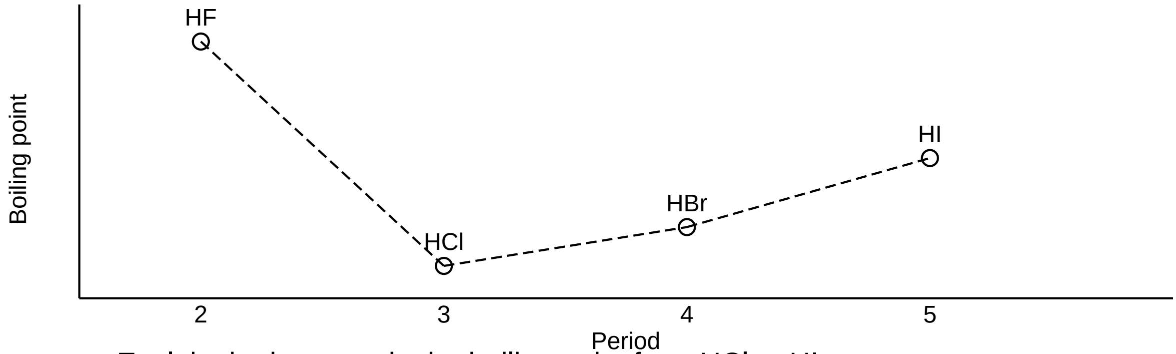
(ii) The graph shows the boiling points of the hydrides of group 17 elements.



Explain the increase in the boiling point from HCl to HI. [2]

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(This question continues on the following page)

(Question 7 continued)

(b) Lewis structures show electron domains and are used to predict molecular geometry.

Deduce the electron domain geometry and the molecular geometry for the PCl_3 molecule. [2]

Electron domain geometry:

. . . .

Molecular geometry:

. . . .

(c) Resonance structures exist when a molecule can be represented by more than one Lewis structure.

(i) Sulfur trioxide, SO_3, can be represented by resonance structures. Calculate the formal charge on each oxygen atom in the structures below. [2]

Structure	I	II
O atom (double bond)		
O atom (single bond)		

(ii) Deduce, giving a reason, the most likely structure of SO_3. [1]

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(d) Absorption of UV light causes dissociation of halogen molecules in the stratosphere. Identify, in terms of bonding, whether Cl_2 or F_2 requires a longer wavelength to dissociate. [2]

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(This question continues on the following page)

(Question 7 continued)

(e) Carbon and sulfur are elements in group 16.

Explain why CO_2 is a gas but CS_2 is a liquid at room temperature. [2]

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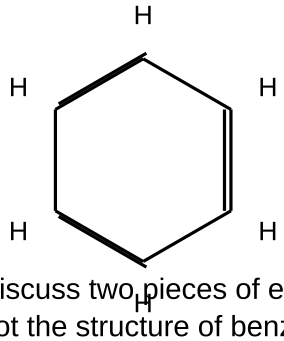
8. The structure of an organic molecule can help predict the type of reaction it can undergo.

(a) The Kekulé structure of benzene suggests it should readily undergo addition reactions.



Discuss two pieces of evidence, one physical and one chemical, which suggest this is not the structure of benzene. [2]

Physical evidence:

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Chemical evidence:

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(This question continues on the following page)

(Question 8 continued)

(b)

(i) Formulate the ionic equation for the oxidation of butan-1-ol to the corresponding aldehyde by acidified dichromate(VI) ions. Use section 24 of the data booklet. [2]

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(ii) The aldehyde can be further oxidized to a carboxylic acid.
Outline how the experimental procedures differ for the synthesis of the aldehyde and the carboxylic acid. [2]

Aldehyde:

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Carboxylic acid:

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(c) Improvements in spectroscopy have made identification of organic compounds more reliable.

The empirical formula of an unknown compound containing a phenyl group was found to be C_3H_3O. The molecular ion peak in its mass spectrum appears at m/z = 102.

(i) Deduce the molecular formula of the compound. [1]

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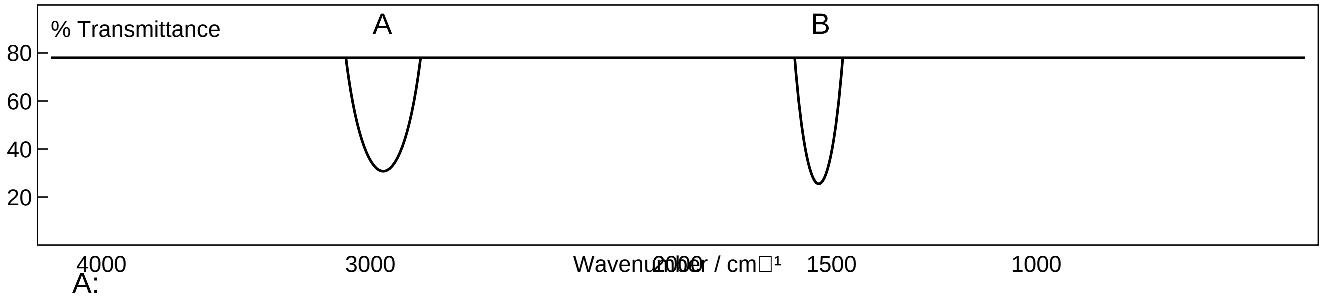
(This question continues on the following page)

(Question 8 continued)

(ii) Identify the bonds causing peaks A and B in the IR spectrum of the unknown compound using section 26 of the data booklet. [1]



A:

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B:

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(iii) Deduce full structural formulas of two possible isomers of the unknown compound, both of which are esters. [2]

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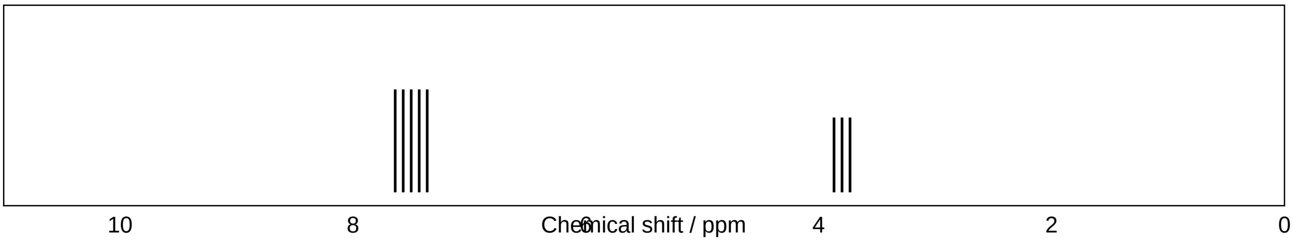
(Question 8 continued)

(iv) Deduce the structural formula of the unknown compound based on its ^1H NMR spectrum using section 27 of the data booklet. [1]



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9.

(a) Organic compounds often have isomers.

A straight chain molecule of formula $C_6H_{12}O$ contains a carbonyl group. The compound cannot be oxidized by acidified potassium dichromate(VI) solution.

(i) Deduce the structural formulas of two possible isomers. [2]

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(This question continues on the following page)

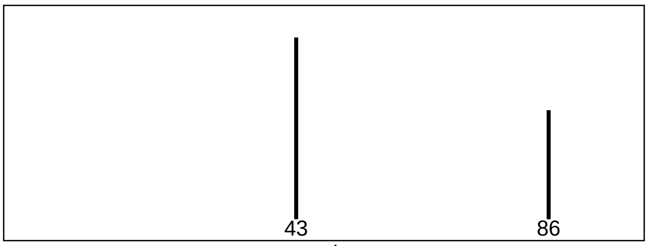
Turn over

(Question 9 continued)

(ii) Mass spectra A and B of the two isomers are given.

Explain which spectrum is produced by each compound using section 28 of the data booklet. [2]

A



m/z

B

m/z

A:

. . . .

. . . .

B:

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(This question continues on the following page)

(Question 9 continued)

(b) A tertiary halogenoalkane, $(CH_3)_2C(C_2H_5)Br$, undergoes an S_N1 reaction with water and forms two stereoisomers.

(i) State the type of bond fission that takes place in an S_N1 reaction. [1]

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(ii) State the type of solvent most suitable for the reaction. [1]

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(iii) Draw the structure of the carbocation intermediate formed, and state its shape. [2]

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Shape:

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(iv) Suggest, giving a reason, the percentage of each stereoisomer from the S_N1 reaction [2]

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(This question continues on the following page)

(Question 9 continued)

(c) Nitrotoluene, $CH_3C_6H_4NO_2$, can be converted to methylphenylamine via a two-stage reaction similar to the conversion of nitrobenzene to phenylamine.

Formulate the equation for each stage of the reaction. [2]

Stage one:

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Stage two:

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